

TOKUSHIMA, Japan

WMO: 478950

Lat: 34.0678N

Lon: 134.5733E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	1.4	-8.6	1.8	4.2	-7.2	2.0	4.7	9.3	6.5	8.5	6.2	3.2	290	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	33.5	25.6	32.4	25.4	31.5	25.2	26.7	31.1	26.3	30.5	25.8	29.9	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	20.9	29.1	25.1	20.3	28.7	24.6	19.7	28.4	84.1	31.2	81.9	30.6	79.9	30.0	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.8	7.6	6.8	-1.4	35.7	1.1	0.9	-2.2	36.4	-2.9	36.9	-3.5	37.4	-4.3	38.1	
			DB												
			WB	-3.5	27.4	1.2	0.5	-4.4	27.8	-5.1	28.2	-5.8	28.5	-6.7	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.3	6.8	9.9	15.0	19.8	23.2	27.2	28.5	25.1	19.6	13.8	8.7	(6)
		DBStd	8.06	2.20	2.60	2.96	2.95	2.13	2.09	2.05	1.66	2.48	2.67	2.92	2.69	
		HDD10.0	313	118	95	39	2	0	0	0	0	0	3	57		
		HDD18.3	1522	374	322	261	104	10	0	0	0	16	137	298		
		CDD10.0	2890	2	6	37	153	303	396	534	452	298	118	17		
		CDD18.3	1058	0	0	0	4	55	146	276	317	202	57	2	0	
		CDH23.3	9107	0	0	0	13	174	714	2774	3681	1579	172	1	0	
		CDH26.7	3011	0	0	0	1	13	114	958	1498	413	15	0	0	
(14)	Wind	WSAvg	3.1	3.4	3.4	3.4	3.3	3.3	2.8	3.0	3.1	3.1	2.9	2.9	3.2	(14)
(15)	Precipitation	PrecAvg	1657	47	65	98	109	151	204	175	188	271	173	105	68	(15)
(16)		PrecMax	2495	113	147	179	226	433	382	404	952	849	469	282	249	(16)
(17)		PrecMin	992	7	14	46	41	45	55	63	22	68	19	15	6	(17)
(18)		PrecStd	410	29	36	37	49	94	76	84	196	176	111	70	52	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.7	18.0	20.6	25.1	28.2	31.6	34.6	35.4	32.8	28.4	22.6	18.2		(19)
(20)		0.4% MCWB	10.7	14.2	14.3	17.1	18.1	24.2	25.5	25.4	25.0	22.7	17.3	14.3		(20)
(21)		2% DB	12.7	15.0	18.5	22.8	26.5	29.1	33.0	33.9	31.4	26.4	21.2	16.0		(21)
(22)		2% MCWB	8.7	11.0	13.4	15.7	17.9	22.5	25.7	25.7	24.3	21.0	16.6	12.1		(22)
(23)		5% DB	11.5	12.9	17.0	21.4	25.3	27.9	32.0	32.9	30.2	25.2	20.0	14.5		(23)
(24)		5% MCWB	7.3	9.0	12.1	15.5	18.0	21.8	25.5	25.5	23.6	20.0	15.8	10.4		(24)
(25)		10% DB	10.2	11.4	15.4	20.1	24.0	26.8	31.0	32.0	29.1	24.0	18.8	13.2		(25)
(26)		10% MCWB	6.2	7.5	11.1	14.5	17.8	21.4	25.2	25.4	23.3	19.1	14.8	9.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	12.2	14.9	16.6	19.7	22.5	25.3	27.2	27.6	26.5	24.5	19.5	15.7		(27)
(28)		0.4% MCDB	13.9	17.5	18.8	22.3	24.9	29.3	31.6	32.0	30.0	26.9	21.0	16.8		(28)
(29)		2% WB	9.7	12.0	14.7	18.1	21.0	24.3	26.6	27.0	25.7	22.7	17.9	12.9		(29)
(30)		2% MCDB	11.8	14.5	17.2	20.7	23.4	27.5	30.9	31.5	29.0	25.2	20.0	15.4		(30)
(31)		5% WB	7.9	9.9	13.2	16.8	20.0	23.6	26.1	26.5	25.1	21.2	16.6	11.1		(31)
(32)		5% MCDB	10.7	12.2	16.1	19.9	22.7	26.4	30.3	30.9	28.4	23.8	19.3	13.8		(32)
(33)		10% WB	6.6	8.0	11.7	15.7	19.0	22.9	25.7	26.1	24.6	20.0	15.3	9.7		(33)
(34)		10% MCDB	9.6	10.7	14.9	18.9	22.2	25.4	29.6	30.4	27.8	23.1	18.2	12.7		(34)
(35)	Mean Daily Temperature Range	MDBR	6.3	6.9	7.8	8.1	7.7	6.2	6.0	6.6	6.2	6.5	6.8	6.6		(35)
(36)		5% DB MCDBR	8.1	9.1	10.0	9.7	9.5	7.7	7.3	7.8	7.2	7.5	7.8	8.2		(36)
(37)		5% DB MCWBR	5.7	6.6	7.0	5.6	4.9	3.4	2.9	3.1	3.3	4.1	5.0	5.7		(37)
(38)		5% WB MCDBR	7.8	8.5	9.2	8.0	7.1	6.0	6.7	7.0	6.0	6.1	7.0	7.8		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.400	0.448	0.509	0.542	0.548	0.563	0.562	0.534	0.497	0.459	0.424	0.394		(40)
(41)		taud	2.170	2.051	1.905	1.878	1.934	1.969	2.019	2.090	2.161	2.185	2.188	2.203		(41)
(42)		Ebn at Noon	781	781	765	762	762	748	746	761	770	764	750	762		(42)
(43)		Edh at Noon	118	148	185	199	190	183	173	159	142	128	115	108		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.58	3.16	4.12	4.87	5.20	4.61	5.12	5.30	4.12	3.36	2.62	2.31		(44)
(45)		RadStd	0.18	0.24	0.29	0.48	0.52	0.47	0.62	0.48	0.36	0.35	0.23	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page